EMOTIX: A WEARABLE AI SYSTEM FOR EMPLOYEE ASSISTANT AND MENTAL WELLNESS

25-26J-007

Project Proposal Report

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B.Sc. (Hons) Degree in Information Technology
Specializing in Computer Systems & Network Engineering

Department of Computer Systems Engineering

Sri Lanka Institute of Information Technology
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DECLARATION

I declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or

Diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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ABSTRACT

Workplace stress is a critical factor affecting employee performance and well-being, yet most existing systems rely on static CNN-only models or insecure architectures that fail to address real-world deployment needs. This research proposes a secure, network-based AI system for real-time stress detection and intervention that integrates machine learning, networking, and data security. The detection framework employs a Hybrid CNN–LSTM model, combining convolutional neural networks for spatial facial feature extraction with long short-term memory networks for temporal stress progression analysis. This approach improves detection accuracy beyond 90% by capturing both immediate facial cues and stress build-up over time.

Once stress is detected, the system transmits results through MQTT communication secured with TLS 1.3, ensuring that only authenticated devices can send data across the network. Stress records are stored in a centralized database encrypted with AES-256 in GCM mode, where each entry uses a unique initialization vector (IV = Device_ID + Counter). This guarantees both confidentiality and integrity of stress data while preserving employee privacy. Importantly, employee histories remain consistent across desks or devices through secure identity association. The system also incorporates a personalized recommendation engine, delivering real-time interventions such as breathing exercises, mindfulness prompts, or short breaks within five seconds of detection. An interactive dashboard visualizes individual stress patterns for employees while presenting only anonymized, aggregated insights into HR and management, ensuring trust and compliance with privacy requirements.

This work contributes to a complete and secure stress detection ecosystem that addresses both accuracy and data protection gaps. By uniting deep learning with secure networking, encryption, and IoT-based deployment, the proposed system enhances workplace well-being while ensuring scalability, privacy, and organizational trust.

Keywords: Stress Detection, CNN–LSTM, TLS 1.3, AES-256 GCM, IVs, Secure Networking, Centralized Database, Personalized Interventions, Workplace Well-being.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
CNN	Convolutional Neural Network
LSTM	Long Short-Term Memory
CNN–LSTM	Hybrid Convolutional Neural Network and Long Short-Term Memory
IoT	Internet of Things
MQTT	Message Queuing Telemetry Transport
REST	Representational State Transfer
API	Application Programming Interface
UI/UX	User Interface / User Experience
PC	Personal Computer
DB	Database
GPU	Graphics Processing Unit
AES	Advanced Encryption Standard
TLS	Transport Layer Security
CI/CD	Continuous Integration / Continuous Deployment

1. INTRODUCTION

1.1 Background

Workplace stress has become one of the most pressing challenges in modern organizations, particularly within the IT industry, where senior engineers face demanding workloads, tight deadlines, and the dual responsibility of managing complex projects alongside personal commitments. The World Health Organization reports that stress-related mental health conditions affect over 264 million people globally, costing the global economy more than $1 trillion annually in lost productivity [1]. In high-pressure IT environments, unmanaged stress cascades into burnout, decreased performance, and long-term health issues, making proactive monitoring and intervention critical for both individual well-being and organizational efficiency.

Traditional stress management approaches, including self-reported surveys and reactive interventions following visible breakdowns, prove insufficient in fast-paced technological environments [2]. These methods fail to capture real-time emotional fluctuations and typically intervene too late to prevent significant productivity losses. The integration of Artificial Intelligence (AI) and computer vision technologies provides a transformative pathway for evolving stress detection from reactive to proactive, real-time monitoring [3], [4]. Facial expression recognition offers continuous, non-intrusive, and objective insights into emotional well-being, as micro-expressions often reveal stress indicators long before individuals consciously acknowledge their psychological state [5].Recent advances have highlighted the importance of integrating edge computing architectures to address privacy and responsiveness requirements. Local analysis on workplace devices ensures that raw video data never leaves the immediate processing environment, preserving employee privacy while enabling low-latency detection essential for timely intervention [6]. However, existing systems exhibit critical limitations: most rely exclusively on CNN-based models that ignore temporal stress progression patterns, and virtually none address the substantial security risks associated with transmitting and storing sensitive biometric data in networked workplace environments.

To bridge these gaps, this research proposes a comprehensive solution integrating a hybrid CNN-LSTM architecture that combines CNN-based spatial facial feature extraction with LSTM-based temporal modeling to track how stress builds and evolves over time. This approach provides more nuanced understanding of stress progression than spatial analysis alone. Additionally, a robust end-to-end security framework protects sensitive workplace data: results are transmitted over TLS 1.3 encrypted channels with mutual authentication (mTLS), while all stored data is protected using AES-256 GCM encryption with unique Initialization Vectors (IVs), ensuring every record remains confidential and tamper-proof throughout its lifecycle.The system incorporates an intelligent recommendation engine that dynamically selects personalized stress interventions from a comprehensive pool including guided breathing

exercises, ambient environmental adjustments, mindfulness prompts, and team-based wellness activities. The algorithm considers individual stress patterns, contextual factors such as time and workload, and historical intervention effectiveness to maximize stress reduction success rates.

By uniting advanced AI-driven temporal stress detection with comprehensive security mechanisms and personalized intervention strategies, this research delivers not only accurate and timely stress monitoring but also establishes a scalable, privacy-preserving solution specifically designed for IT workplace environments. The anticipated outcome is a comprehensive system capable of simultaneously improving employee well-being and enhancing organizational productivity through secure, personalized stress interventions that maintain the highest standards of data protection while addressing individual needs in real-time.

1.2 Literature Review

1.2.1 Traditional Stress Detection Methods

Traditional stress detection has largely relied on self-report questionnaires, periodic surveys, and physiological measurements. While valuable, these approaches require conscious user input and are prone to bias including social desirability effects and recall errors, alongside delayed recognition of symptoms [2], [6]. Tools such as the Perceived Stress Scale and Depression Anxiety Stress Scales depend on individuals accurately reporting their emotional state, usually after stress has already intensified. Physiological monitoring via heart rate variability, galvanic skin response, or cortisol levels shows promise in laboratory settings but faces significant challenges in real-world workplaces due to specialized sensor requirements, invasive procedures, and impractical sample collection methods. Moreover, these physiological signals serve as indirect stress indicators and can be substantially confounded by non-stress factors including physical activity, caffeine intake, or underlying health conditions, leading to false positives. The inherently reactive nature of traditional methods means interventions typically occur only after stress becomes severe, missing critical opportunities for early prevention and support [6].

1.2.2 Convolutional Neural Networks in Facial Expression Recognition

Computer vision and deep learning have introduced transformative methods for stress detection, with Convolutional Neural Networks (CNNs) proving highly effective in facial expression recognition [4], [8]. CNNs automatically extract spatial features including facial edges, textures, and muscle action patterns correlated with emotional states, enabling detection of subtle facial movements linked to stress, anxiety, frustration, or fatigue [9]. Their hierarchical, layered learning structure transforms raw pixel data into increasingly abstract emotional representations, providing objective and non-intrusive insights into psychological well-being [10]. However, standard CNN-based approaches typically analyze static images, fundamentally limiting their ability to capture emotional transitions or stress accumulation over time. This temporal limitation

creates a significant blind spot that reduces effectiveness in continuous monitoring scenarios where understanding stress progression patterns is crucial for effective intervention

1.2.3 LSTM Networks for Temporal Pattern Analysis

Long Short-Term Memory (LSTM) networks address temporal limitations by processing sequential data and retaining memory of prior states. Their ability to capture long-term dependencies makes them particularly well-suited for analyzing emotional changes across extended time periods. Research demonstrates that LSTM-based models can identify early indicators of rising stress and predict stress escalation by analyzing sequences of emotional data [7]. This predictive capacity enables proactive intervention strategies, where warning signs of accumulating stress trigger coping prompts before symptoms worsen. However, LSTMs alone struggle with extracting rich spatial features from raw facial inputs, often requiring pre-processed feature sequences such as facial landmarks or CNN-learned features, which limits real-world applicability where end-to-end feature learning is essential [13].

1.2.4 Hybrid CNN-LSTM Architectures

Given the complementary strengths and limitations of CNNs and LSTMs, recent literature has increasingly focused on hybrid CNN-LSTM architectures. The CNN-LSTM approach integrates CNN-based spatial feature extraction with LSTM-based temporal modeling to capture both detailed facial characteristics and their evolution over time [9]. In such systems, the CNN component first learns facial expression features including muscle movements and Action Units from each video frame, with these features subsequently fed into the LSTM which learns temporal dynamics across frame sequences [10]. This combined architecture can detect subtle stress-related expressions while tracking their progression and fluctuation over time, yielding more accurate and robust stress detection capabilities than either model architecture alone. However, existing implementations lack optimization for edge deployment and fail to integrate with comprehensive security frameworks necessary for workplace environments.

1.2.5 Personalized Intervention Systems and Alert Mechanisms.

Research on personalized stress intervention systems remains significantly limited, highlighting a critical gap in current stress management technology. Most existing digital stress management tools apply generic, one-size-fits-all coping suggestions without accounting for individual effectiveness, contextual factors, or predictive stress indicators [2]. These rule-based responses ignore crucial factors including personal stress triggers, historical coping success rates, and current environmental conditions. While emerging systems have begun exploring just-in-time adaptive interventions using sensor data and AI to tailor timing and content of support, such approaches remain in early developmental stages [8]. Few workplace platforms integrate truly adaptive recommendation engines that analyze individual employee stress histories, real-time physiological and emotional data, and situational context to automatically select optimal interventions from pools of breathing exercises, mindfulness prompts, calming music, and break

reminders [7], [9]. Furthermore, most systems operate on scheduled check-ins or user-initiated help requests, often delivering notifications after stress has peaked rather than providing real-time, preventative alerts within critical time windows (≤5 seconds) [14].

1.2.6 Local Stress Detection & Metadata Generation.

Edge-based stress detection represents a critical advancement in privacy-preserving biometric monitoring, yet current implementations suffer from significant limitations in both processing capabilities and data handling approaches. Existing edge computing solutions for emotion recognition typically process raw facial images locally but often transmit processed features or even raw data to external servers for final classification, compromising privacy objectives [11]. The challenge of deploying sophisticated hybrid models on resource-constrained devices has led most researchers to implement simplified CNN architectures that sacrifice temporal analysis capabilities [12]. Current systems lack comprehensive metadata generation frameworks that can extract meaningful stress indicators while preserving privacy through exclusive local processing. The absence of standardized metadata schemas including Employee_ID, Stress_Level classifications (low/moderate/high), quantitative Stress_Score metrics, precise Timestamp recording, and Device_ID tracking limits the effectiveness of longitudinal stress monitoring and cross-device data correlation [13]. Moreover, existing edge implementations fail to guarantee that raw video data never leaves the local device, creating potential privacy vulnerabilities in workplace environments where biometric data protection is paramount [14].

1.2.7 Secure Communication with MQTT and Centralized Database Security.

The integration of secure communication protocols in biometric monitoring systems remains inadequately addressed in current literature, with most existing stress detection platforms lacking comprehensive end-to-end security architectures. Traditional emotion recognition systems often rely on outdated communication protocols, basic MQTT without encryption, creating substantial vulnerabilities during data transmission between edge devices and centralized systems [15]. While some research has explored basic TLS implementations, few studies have investigated the application of modern TLS 1.3 protocols with mutual authentication (mTLS) specifically for biometric stress data transmission [16]. Current MQTT-based implementations typically lack robust device authentication mechanisms, allowing potentially unauthorized devices to inject malicious data or intercept legitimate communications [17].

Centralized database security for biometric data presents even more significant challenges, with existing systems typically implementing basic AES encryption without proper initialization vector (IV) management or integrity protection mechanisms. The critical security requirement for unique IVs in AES-256 GCM mode to prevent key-IV pair reuse and subsequent cryptographic attacks remains unaddressed in current stress monitoring implementations [18]. The absence of comprehensive nonce generation strategies using Device_ID and Counter combinations leaves stored stress records vulnerable to pattern analysis attacks and data

tampering [19]. Furthermore, most existing systems fail to implement proper confidentiality and integrity guarantees simultaneously, with GCM authentication tags rarely utilized to ensure tamper detection in stored stress data [20]. This comprehensive security gap represents a fundamental barrier to enterprise deployment of biometric stress monitoring systems where regulatory compliance and data protection are essential requirements [21].

1.3 Research Gap

Stress detection and mental health monitoring in workplace environments present unique challenges that current technologies have not fully addressed, particularly in real-time, privacy-preserving, and secure deployment settings. Traditional approaches, which primarily rely on self-reporting mechanisms and physiological sensors, are reactive in nature and often miss critical opportunities for early intervention. Although advances in computer vision and machine learning have led to the development of AI-based emotion recognition systems, these solutions are not without significant limitations when deployed in enterprise environments.

Recent research initiatives have demonstrated how deep learning techniques can be applied to facial expression recognition for stress detection. However, these approaches are designed for quite specific use cases and fail to address the broader requirements for comprehensive stress management systems that can operate in real-world workplace environments while providing immediate, personalized interventions. A fundamental limitation lies in the widespread reliance on CNN-only architectures, which capture spatial facial features but completely ignore temporal stress progression patterns. This temporal blindness prevents systems from understanding how stress builds, fluctuates, and evolves over time, resulting in incomplete emotional analysis that misses critical stress escalation indicators.

Furthermore, while individual deep learning architectures perform well in controlled laboratory situations, they frequently lose effectiveness when subjected to the complex temporal dynamics, networking requirements, and security constraints found in practical workplace applications. Most existing systems either process data exclusively on local devices—severely limiting scalability and cross-device stress tracking—or transmit sensitive biometric information to cloud servers without adequate cryptographic protection, creating substantial privacy vulnerabilities and trust barriers for enterprise adoption.

Moreover, the majority of existing emotion recognition systems lack the integration necessary for real-time stress detection coupled with intelligent, personalized intervention delivery. Current solutions typically stop at detection or provide generic, one-size-fits-all recommendations without considering individual user contexts, historical effectiveness, or dynamic environmental factors. No existing approach addresses the critical need for secure end-to-end communication protocols (such as TLS 1.3 with mutual authentication) combined with centralized encrypted storage systems (utilizing AES-256 GCM with unique initialization vectors) that can protect

sensitive stress data while enabling comprehensive longitudinal analysis and employee mobility across workplace devices.

This comprehensive gap emphasizes the necessity for an innovative solution that leverages the latest developments in hybrid CNN-LSTM architectures for spatio-temporal analysis, secure networking protocols for enterprise-grade data protection, centralized encrypted databases for scalable storage, and personalized recommendation systems while adapting to the unique demands of IT workplace mental health monitoring. Such a system would enhance both the accuracy and security of stress detection while providing immediate, context-aware, and privacy-preserving support to prevent stress escalation in high-pressure professional environments.

Comparison with Existing Research

Research A [Kumar et al., 2025]: Focuses on deep CNN models for stress detection through facial expression recognition, achieving high accuracy in spatial feature extraction but lacking temporal understanding of stress progression patterns, personalized intervention capabilities, and secure communication protocols for enterprise deployment.

Research B [Babu et al., 2025]: Addresses real-time stress detection using integrated facial expression analysis with hybrid neural networks, successfully combining CNN with other methods but limited compared to CNN-LSTM spatio-temporal modeling, without intelligent intervention selection, and lacking secure end-to-end communication and encrypted storage mechanisms.

Research C [Yadav, 2024]: Implements real-time stress detection using facial images with CNNs, demonstrating edge computing capabilities for privacy preservation but limited to simple CNN architecture without temporal analysis, personalized recommendation systems, or secure networking protocols for centralized data management.

Research D [Almeida & Rodrigues, 2024]: Develops CNN combined with Haar cascade techniques for real-time emotion-based stress detection in AI systems, achieving effective detection capabilities but lacking temporal modeling, comprehensive intervention delivery mechanisms, and enterprise-grade security for data transmission and storage.

Table 1 : Research Gap

Research	CNN-Based Spatial Analysis	LSTM-Based Temporal Analysis	Hybrid Architecture	Personalized Intervention Selection	Accuracy >90%	Secure Communication (TLS 1.3 in transit)	Centralized DB with AES-256 GCM + IVs (Encryption)

Research A [Kumar et al., 2025]	✓	✗	✗	✗	✓	✗	✗
Research B [Babu et al., 2025]	✓	✗	**Limited**	✗	✓	✗	✗
Research C [Yadav, 2024]	✓	✗	✗	✗	✗	✗	✗
Research D [Almeida & Rodrigues, 2024]	✓	✗	✗	✗	✗	✗	✗
Proposed System (Your Work)	✓	✓	✓	✓	✓	✓	✓

*Limited = Basic hybrid approach without CNN-LSTM spatio-temporal modeling.

The analysis reveals three fundamental limitations across existing research that prevent enterprise-ready deployment:

Temporal Modeling Gap: No existing system [Kumar et al., Babu et al., Yadav, Almeida & Rodrigues] combines CNN spatial feature extraction with LSTM temporal analysis for comprehensive spatio-temporal stress understanding. While Research A and Research B demonstrate strong CNN-based spatial analysis, none achieve true temporal modeling that captures stress progression patterns over time, missing critical opportunities for predictive intervention.

Security Architecture Gap: Most critically, all reviewed systems completely lack secure communication protocols and centralized encrypted storage mechanisms essential for workplace deployment. Research C attempts edge computing for privacy but fails to address secure data transmission, while others ignore networking security entirely. No existing approach implements

TLS 1.3 for secure communication or AES-256 GCM with unique initialization vectors for tamper-proof centralized storage.

Intervention Integration Gap: The absence of dynamic, personalized intervention selection is evident across all systems. Research C and Research D focus solely on detection without intervention capabilities, while others provide generic recommendations without intelligent algorithms that predict intervention effectiveness based on individual patterns, contextual factors, and historical success rates.

These comprehensive gaps highlight the necessity for an integrated solution that unifies accurate spatio-temporal detection, enterprise-grade security, and personalized intervention delivery. The proposed system directly addresses these limitations by combining CNN-LSTM hybrid architecture with secure networking protocols and encrypted centralized storage, creating the first complete stress management ecosystem suitable for real workplace environments

1.4 Research Problem

Work-related stress represents one of the most pressing challenges in modern professional environments, particularly within high-pressure IT workplaces where senior engineers face complex project demands, tight deadlines, and continuous technological evolution. This significantly impacts employee health, productivity, and organizational sustainability. Traditional stress detection methods, such as self-report surveys and periodic assessments, are reactive, subjective, and incapable of capturing real-time emotional fluctuations. As a result, opportunities for early intervention are often missed, leading to escalating stress, burnout, and reduced workplace performance.

Recent advances in artificial intelligence (AI), computer vision, and edge computing provide a pathway toward proactive, real-time stress monitoring systems that can detect early signs of stress and deliver personalized interventions. However, existing approaches remain fundamentally limited in five critical dimensions: (1) over-reliance on static CNN models that ignore temporal dynamics of stress progression, (2) lack of personalization to account for individual emotional variability, (3) reactive rather than proactive intervention mechanisms, (4) absence of secure communication protocols for enterprise deployment, and (5) inadequate centralized data protection mechanisms that compromise trust and scalability.

First, most current systems rely heavily on CNN-based spatial analysis, which can detect facial features and micro-expressions but fail to capture the temporal evolution of stress. Stress develops through subtle, progressive changes over time, requiring spatio-temporal modeling to anticipate escalating levels. While some hybrid models exist, they remain limited and do not achieve comprehensive CNN-LSTM integration that could provide robust temporal stress progression analysis.

Second, existing frameworks lack personalization, adopting one-size-fits-all approaches that ignore individual stress expression patterns, cultural backgrounds, and baseline emotional states. This leads to frequent false positives and negatives. Moreover, interventions are typically generic and static, offering basic reminders or relaxation prompts without adapting dynamically to user

history, preferences, or contextual effectiveness. The absence of intelligent recommendation engines capable of selecting optimal, context-aware interventions in real time remains a critical limitation.

Third, most stress detection systems operate reactively, identifying stress only after escalation and offering limited coping strategies that fail to prevent performance impact. The detection-to-intervention gap persists, where systems may detect stress accurately but rarely connect this to timely, adaptive support at the exact moment of need.

Fourth, secure communication protocols are absent from existing research, creating significant barriers to enterprise adoption. Current systems either process data exclusively on local devices limiting scalability and cross-device tracking or transmit sensitive biometric information without adequate protection. No existing approach implements TLS 1.3 with mutual authentication to ensure only trusted devices can transmit stress data, leaving workplace deployments vulnerable to security breaches.

Finally, centralized data protection remains unaddressed, preventing secure longitudinal analysis and employee mobility across workplace devices. Existing systems lack encrypted storage mechanisms such as AES-256 GCM with unique initialization vectors, which are essential for maintaining both confidentiality and integrity of sensitive stress records while enabling comprehensive organizational insights.

In summary, while existing research has advanced stress detection through various CNN architectures and basic hybrid models, the field remains fragmented, with solutions focusing on isolated components rather than comprehensive integration. The core research problem addressed in this study is the design and implementation of a secure, AI-powered, real-time stress detection and intervention ecosystem that unifies CNN-LSTM spatio-temporal emotion recognition, personalized adaptive interventions, proactive stress management, secure end-to-end communication protocols, and encrypted centralized storage. This comprehensive framework seeks to bridge multiple gaps simultaneously—detection accuracy, personalization, security, and scalability—enabling IT workplace environments to provide the right support at the right moment while maintaining enterprise-grade data protection and employee trust.

2. OBJECTIVES

2.1 Main Objectives

The main objective of this research is to design, develop, and evaluate a secure, AI-powered real-time stress detection and intervention system for IT workplace environments that provide accurate, privacy-preserving, and context-aware support for employees while ensuring enterprise-grade data protection. Unlike traditional stress management approaches that are reactive, subjective, and generic, the proposed system aims to establish a comprehensive closed-loop framework that continuously monitors employees' emotional states using hybrid Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) models to capture both spatial facial features and temporal stress progression patterns.

The system integrates edge computing architectures to ensure low-latency, real-time operation while safeguarding user privacy by processing facial data locally and transmitting only stress metadata. Critical to enterprise deployment, the framework implements secure end-to-end communication protocols using TLS 1.3 with mutual authentication to guarantee that only trusted devices can transmit stress data, while a centralized database encrypted with AES-256 in GCM mode using unique initialization vectors ensures both confidentiality and integrity of stored stress records. This secure architecture enables employee stress histories to follow them across different workplace devices while maintaining complete data protection. The system delivers personalized, just-in-time adaptive interventions through an intelligent recommendation engine that selects optimal coping strategies including breathing exercises, mindfulness prompts, calming music, and break reminders, based on individual user contexts, preferences, and historical effectiveness within critical response timeframes of five seconds or less.

By bridging the gap between accurate spatio-temporal stress detection, secure networking protocols, encrypted centralized storage, and actionable personalized intervention, this research seeks to transform workplace stress management from a reactive paradigm into a proactive, intelligent, secure, and user-centered ecosystem. The comprehensive framework aims to improve employee well-being, enhance productivity, reduce the risk of burnout, while establishing the trust and security standards necessary for successful enterprise adoption in high-pressure IT workplace environments.

2.2 Specific Objectives

The proposed research is guided by several specific objectives that collectively ensure the development of a robust, intelligent, secure, and practical stress detection and intervention system for IT workplace environments.

1. Develop a hybrid CNN–LSTM stress detection model. The first objective is to construct and train a deep learning architecture that integrates Convolutional Neural Networks (CNNs) for spatial feature extraction with Long Short-Term Memory (LSTM) networks for temporal sequence modeling. CNNs will capture fine-grained spatial cues such as micro-expressions and facial muscle movements, while LSTMs will analyze how these expressions evolve over time to detect stress progression patterns. This dual modeling approach addresses the limitations of CNN-only static analysis and LSTM-only sequential analysis, enabling more accurate detection of subtle and progressive stress signals. The proposed model will be benchmarked against baseline CNN and LSTM systems to demonstrate improvements in spatio-temporal accuracy beyond 90%.

2. Implement secure end-to-end communication protocols. A critical objective is to establish enterprise-grade security for stress data transmission between workplace devices and centralized systems. The system will implement TLS 1.3 with mutual authentication (mTLS) to ensure only trusted, authenticated devices can transmit stress metadata. This secure communication framework will protect sensitive workplace stress information during transit while enabling real-time alert delivery and centralized trend analysis essential for organizational well-being insights.

3. Design encrypted centralized database architecture. To enable secure longitudinal stress analysis and employee mobility across workplace devices, the system will implement a centralized database encrypted with AES-256 in Galois/Counter Mode (GCM). Each stress record will use a unique initialization vector (IV) generated from Device_ID and Counter combinations, ensuring both confidentiality and integrity protection. This architecture will allow employee stress histories to follow them across different workplace setups while maintaining complete data protection and tamper detection capabilities.

4. Develop intelligent recommendation and intervention engines. Beyond detection, the system must actively support employees through adaptive interventions. An intelligent recommendation engine will be developed that selects interventions from a diverse pool of evidence-based strategies, including guided breathing exercises, mindfulness prompts, calming music, and break reminders. The selection will rely on predictive analysis using historical user data, contextual information, and stress severity levels to dynamically recommend the most effective intervention for each situation within five seconds of detection.

5. Enhance personalization and adaptability of interventions. This research addresses the limitation of generic stress management by incorporating personalization models that learn each user's unique stress response patterns over time. Factors such as personal coping mechanisms, intervention effectiveness history, and contextual preferences will guide selection, ensuring recommendations are perceived as supportive rather than intrusive. The system will deliver context-aware prompts at appropriate times, fostering acceptance and effectiveness of interventions.

6. Validate system robustness and security in workplace conditions. To ensure practical applicability, the system will be rigorously tested in real IT workplace settings where environmental factors such as varying lighting conditions, multitasking activities, and demographic diversity can challenge recognition accuracy. Additionally, security protocols will be validated to ensure data protection under real network conditions. Unlike models evaluated solely in controlled laboratory conditions, this research emphasizes both accuracy and security resilience in unpredictable office environments.

7. Assess user trust, privacy satisfaction, and adoption potential. Finally, this research aims to investigate employee perceptions of the system's security, transparency, usefulness, and privacy safeguards. Trust is critical for adoption, and employees must feel confident that their stress data is protected through robust encryption and secure communication protocols. User-friendly interfaces, clear communication about data handling, and transparent security measures will be incorporated to ensure long-term adoption, making it a sustainable tool for workplace stress management rather than a short-lived pilot project.

3. METHODOLOGY

This research study employs a comprehensive methodology that leverages cutting-edge deep learning, computer vision, secure networking protocols, and cryptographic techniques to develop an AI-powered real-time stress detection and intervention system for IT workplace environments. To achieve the research objectives of creating a secure, accurate, and personalized stress management ecosystem, a systematic methodological approach will be adopted, encompassing the following key phases:

3.1 System Overview

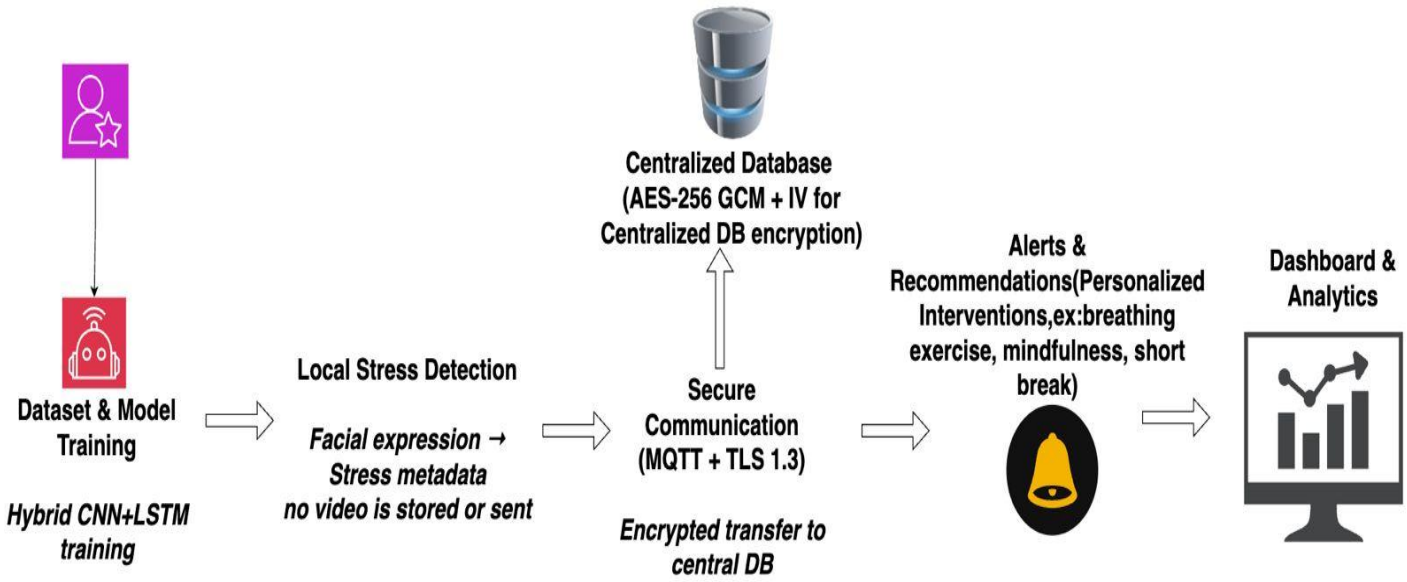

Figure 1 : System Overview

3.2 Requirements Gathering and Analysis

3.2.1 Literature Review

A comprehensive review of prior research on stress detection, emotion recognition, and AI-driven well-being systems [4, 5, 9, 12, 14] was conducted to establish technical requirements and identify limitations in existing solutions. Key gaps observed include the absence of spatio-temporal analysis (most systems rely on CNN-only models), lack of personalization in interventions, and insufficient attention to security mechanisms for transmitting and storing sensitive biometric data. Insights from the literature guided the choice of a Hybrid CNN–LSTM

architecture for spatio-temporal modeling, along with the integration of TLS 1.3-secured communication and AES-256 GCM with unique IVs for enterprise-grade data protection.

3.2.2 Industry Consultations

Structured interviews and focus group discussions were conducted with senior IT engineers, HR managers, and occupational psychologists. IT professionals were prioritized as the primary user group, as they regularly face high workloads, strict deadlines, and the stress of managing complex projects. Consultations provided insights into common workplace stress triggers, acceptable intervention types (e.g., short breaks, mindfulness prompts, breathing exercises), and strong concerns regarding privacy, trust, and surveillance. HR managers emphasized the need for anonymized analytics at an organizational level, while employees expressed preference for interventions that are personalized but non-intrusive.

3.2.3 Practical Observation

Direct observation of IT workplace environments was performed to capture practical deployment challenges, including variable lighting conditions, camera positioning, multitasking scenarios, and dynamic employee interactions. These observations informed requirements for robust real-time stress detection under natural conditions, ensuring the system performs reliably without disrupting workflows. Additionally, monitoring day-to-day activities helped validate the context-sensitive design of the intervention engine, confirming that alerts and recommendations should be delivered within ≤5 seconds but in a manner that does not interrupt critical tasks such as meetings or coding sessions.

3.2.4 Regulatory and Ethical Requirements

The system design was aligned with data privacy regulations, IT industry security policies, and ethical frameworks governing employee monitoring. Key requirements include:

- Local processing on edge devices to ensure no raw video data is transmitted or stored externally.
- Encryption of all records using AES-256 GCM with unique IVs, ensuring confidentiality and integrity of stress histories.
- Secure communication protocols (TLS 1.3 with mTLS) to authenticate devices and prevent unauthorized data access.
- Role-based access control (RBAC) to restrict dashboards and analytics to authorized personnel only, with managers accessing only anonymized reports.
- Employee consent and transparency, ensuring trust and voluntary participation in stress monitoring programs.

3.3 Data Collection

The data collection process for this research is designed to ensure ecological validity, reliability, and privacy preservation while capturing stress-related facial expressions and temporal dynamics in real workplace environments. Data will be collected entirely from employees using Raspberry

Pi 4B units equipped with 8MP camera modules, ensuring that all processing and storage remain local to maintain privacy.

Data Collection Strategy

Data collection in this research focuses entirely on creating a custom dataset from real workplace environments using Raspberry Pi 4B units equipped with 8MP cameras.

- **Facial Expression Data**: Employees will be recorded during regular work activities under varied conditions (lighting changes, camera angles up to 30°, and different office settings). This ensures the dataset reflects natural, real-world variability rather than controlled laboratory conditions.
- **Temporal Sequence Data**: Longer recording sessions (45–60 minutes) will be conducted to capture full stress episodes, from onset to escalation and recovery. These sessions will be segmented into 30–60 second clips at 20 fps, allowing analysis of gradual build-up and acute spikes in stress.
- **Intervention Response Data**: During and after system-generated interventions (e.g., breathing prompts, mindfulness cues, or break reminders), facial expressions and behavioral responses will be recorded. This enables the creation of an intervention-effectiveness dataset that links stress cues with adaptive coping strategies.

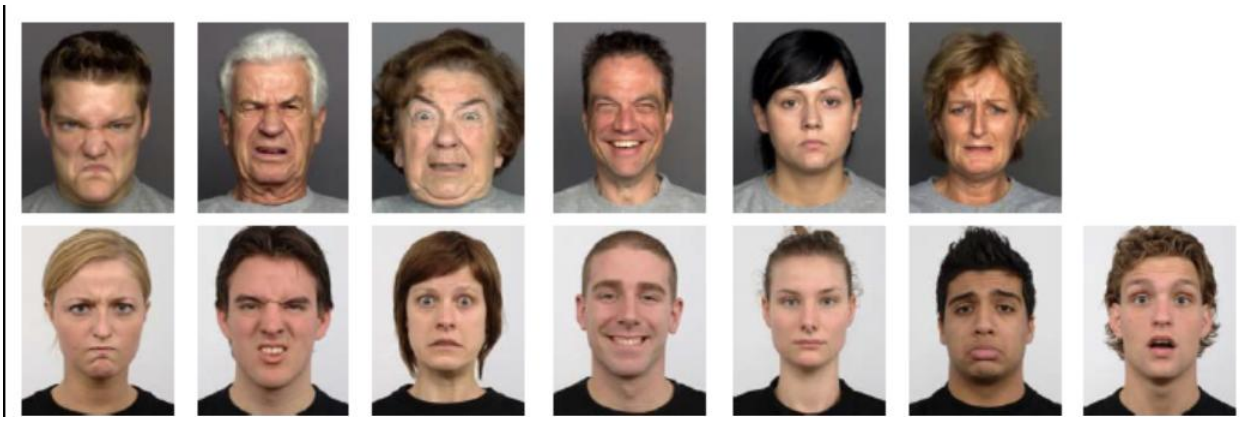

Figure 2 : Emotions of data collection [15]

3.4 Data Preprocessing

3.4.1 Data Preprocessing

Face Detection and Alignment
Faces are detected using OpenCV DNN-based face detectors optimized for workplace variability. Once detected, dlib's 68-point landmark predictor is applied for geometric

normalization of eyes, nose, and mouth, ensuring consistent alignment across frames. This guarantees that the CNN extracts features from properly standardized facial inputs.

Image Enhancement
OpenCV preprocessing improves image quality under diverse workplace lighting conditions. CLAHE (Contrast Limited Adaptive Histogram Equalization) corrects illumination differences, bicubic interpolation resizes frames to 224×224 pixels, and Gaussian filtering reduces noise while preserving fine details such as wrinkles or eye strain indicators, which are critical for stress detection.

Temporal Sequence Preparation
Video streams are standardized to 20 fps and segmented into overlapping sliding windows. Where detection fails, missing frames are interpolated with optical-flow estimation to preserve sequence continuity. This ensures the CNN captures spatial features, while the LSTM models stress progression over time, enabling robust spatio-temporal analysis.

Privacy Preprocessing
All preprocessing is carried out locally on the device, ensuring no raw video leaves the system. Instead, only stress-related metadata (Employee_ID, Stress_Level, Stress_Score, Timestamp, Device_ID) is generated. Before transmission, results are secured using TLS 1.3 with mutual authentication (mTLS) for communication. Stored records in the centralized database are protected using AES-256 in GCM mode with unique IVs (Device_ID + Counter), ensuring both confidentiality and integrity of stress data at rest.

Dataset Splitting
To support training and evaluation, the final dataset is divided into training (70%), validation (15%), and testing (15%) subsets. This ensures robust performance evaluation and fair model comparisons.

3.4.2 Quality Assurance

Annotation Control
Expert annotators, including psychologists, assign stress labels with an inter-rater agreement threshold of $\kappa \geq 0.85$. Temporal continuity checks are performed to avoid inconsistent labeling, and where possible, annotations are cross validated against self-reported stress levels or physiological measures.

Dataset Validation
Systematic checks are carried out to ensure balanced stress distributions (30% low, 40% moderate, 30% high), demographic representation, and dataset integrity. Pilot CNN–LSTM training is used as a benchmark test to confirm annotation quality and dataset usability. Iterative refinement continues until the dataset consistently supports robust stress detection and intervention modeling.

3.5 Hybrid CNN-LSTM Architecture Development

3.5.1 CNN Component Design

Spatial Feature Extraction Architecture:

- Base CNN architecture: Modified ResNet-50 for facial feature extraction
- Custom convolutional layers for micro-expression detection
- Attention mechanisms for focusing on stress-relevant facial regions
- Feature map dimension: 2048 features per frame

3.5.2 LSTM Component Design

Temporal Pattern Analysis Architecture:

- Bidirectional LSTM with 256 hidden units
- Attention-based temporal modeling for stress progression analysis
- Dropout regularization (0.3) to prevent overfitting
- Sequence length: 30 frames (approximately 5-10 seconds)

3.5.3 Hybrid Integration Strategy

- CNN feature extraction followed by LSTM temporal modeling
- Feature fusion techniques for combining spatial and temporal information
- Multi-task learning for stress level classification and progression prediction

3.6 Intelligent Recommendation Engine Development

3.6.1 Intervention Pool Creation

A structured intervention pool will be developed, covering five evidence-based categories: breathing exercises (4-7-8, box breathing, diaphragmatic breathing), mindfulness prompts (body scan, gratitude practice, present-moment awareness), music therapy (classical, nature sounds, binaural beats), physical interventions (stretching, posture correction, short breaks), and cognitive techniques (positive affirmations, reframing exercises). This diversity ensures flexibility and scientific grounding to support varied stress profiles.

3.6.2 Recommendation Algorithm Design

The recommendation algorithm will use a multi-factor decision framework that considers stress severity (low, moderate, high), historical intervention effectiveness, contextual factors such as workload or meeting schedules, user preferences and feedback history, and intervention characteristics like duration and level of disruption. This ensures interventions are context-aware, minimally disruptive, and optimally matched to the employee's current situation.

3.6.3 Personalization Engine

Personalization will be achieved through a combination of collaborative filtering to identify effective strategies across similar users, content-based filtering to match interventions with individual coping styles, reinforcement learning to improve recommendations dynamically through real-time feedback, and A/B testing frameworks to evaluate alternative strategies. Over time, this engine evolves into an adaptive, intelligent system that delivers increasingly personalized and effective interventions.

3.7 Real-Time Communication and Alert System

3.7.1 Communication Protocol Implementation

The system will employ a lightweight MQTT-based architecture for transmitting real-time stress detection outputs. Stress-level updates will be published to topic-based channels (e.g., low, moderate, high stress), allowing targeted delivery. Quality of Service (QoS) level 1 will ensure guaranteed message delivery, while TLS encryption secures all communications, preserving data confidentiality during transmission between Raspberry Pi devices and end-user applications.

3.7.2 Alert Delivery Mechanisms

Alerts will be designed to maximize support without disrupting productivity. Delivery mechanisms include:

- Push notifications to mobile and desktop applications for immediate awareness.
- Escalation protocols, where persistent high-stress levels trigger additional interventions or notify support personnel (in anonymized form).
- Context-aware delivery, ensuring alerts are timed to avoid critical activities like scheduled meetings or coding sessions.
- Multi-modal notifications (visual pop-ups, gentle audio cues, or haptic feedback for wearables), giving employees flexibility to choose their preferred method.

3.8 Security Framework Implementation.

The security framework represents a critical component of the proposed system, addressing enterprise-grade data protection requirements for biometric stress monitoring in workplace environments. This comprehensive security architecture ensures both data confidentiality and integrity throughout the entire system lifecycle, from edge processing to centralized storage and analytics.

3.8.1 Secure Communication Protocols.

TLS 1.3 Implementation: All communication between edge devices and the central system utilizes TLS 1.3, the latest transport security protocol providing enhanced cryptographic strength

and reduced handshake latency. The implementation includes perfect forward secrecy to ensure that even if long-term keys are compromised, past communications remain secure.

MQTT Security Integration: Stress metadata transmission employs MQTT over TLS 1.3 with Quality of Service (QoS) level 1 to guarantee message delivery while maintaining security. Topic-based access control restricts device publishing permissions to authorized stress-level channels. Message payload encryption provides an additional layer of protection beyond transport-level security.

3.8.2 Centralized Database Encryption

AES-256 GCM Implementation: The centralized database implements Advanced Encryption Standard 256-bit encryption in Galois/Counter Mode (GCM) for all stored stress records. GCM mode provides both confidentiality through encryption and integrity protection through built-in authentication tags, ensuring stored data cannot be modified without detection.

Unique Initialization Vector Strategy: Each database record utilizes a unique Initialization Vector (IV) generated through the combination of Device_ID and an incrementing Counter value. This approach ensures IV uniqueness across all records while maintaining deterministic generation for system efficiency. The IV generation formula (IV = Device_ID || Counter) prevents cryptographic vulnerabilities associated with IV reuse while enabling scalable enterprise deployment.

Key Management Architecture: Encryption keys are managed through a dedicated Hardware Security Module (HSM) integration, providing secure key generation, storage, and rotation capabilities. Master keys rotate annually, with automated re-encryption of stored data using new keys while maintaining system availability during rotation periods.

3.9 Dashboard Visualization

The system includes a dashboard that automatically records each stress detection event and corresponding intervention. For every instance, the dashboard logs the time of detection, the stress level, the suggested intervention, and any feedback or outcome. This history view allows employees to monitor their own stress patterns over time, while organizations (with anonymized access) can track broader trends. Visual elements such as time-series charts, intervention effectiveness summaries, and weekly reports make it easier to identify recurring stress periods and evaluate which strategies work best. Beyond simple logging, the dashboard serves as an adaptive insight tool, feeding back into the recommendation engine to improve future personalization.

3.10 Development Methodology

The development of the proposed system will follow an Agile SCRUM framework to ensure iterative progress, continuous feedback, and adaptability to real-world requirements. This approach allows close collaboration with stakeholders (employees, HR professionals, and occupational psychologists) while maintaining flexibility in refining system features.

3.10.1 Agile SCRUM Framework

The project will be structured into 2-week sprints, each delivering functional increments of the system.

- Daily stand-up meetings will be held for quick progress updates and issue resolution.
- Sprint planning and retrospectives will help define tasks, evaluate progress, and refine workflows.
- Continuous integration and deployment (CI/CD) pipelines will ensure seamless model updates and system integration.
- Regular stakeholder feedback will be incorporated, especially regarding usability, privacy, and intervention effectiveness, to keep the solution user-centered.

3.10.2 Development Phases

Model Development and Training (Months 1–4)

- Implementation of the CNN model for baseline feature extraction and later extension to a hybrid CNN–LSTM for spatio-temporal stress analysis.
- Data preprocessing (face alignment, enhancement, anonymization) and training using the custom workplace dataset.
- Edge optimization (e.g., lightweight model compression, quantization) for Raspberry Pi deployment with low latency and strict privacy.
- Initial model validation against annotated datasets to establish baseline accuracy.

System Integration and Testing (Months 5–8)

- Development of the recommendation engine using multi-factor decision criteria and reinforcement learning for personalization.
- Implementation of the communication framework using MQTT and secure TLS protocols for real-time stress alerts.

- Integration of the dashboard visualization module, enabling automated logging of stress events, interventions, and outcomes.
- End-to-end system testing in controlled workplace-like environments to validate functionality.

Field Testing and Validation (Months 9–12)

- Deployment of the system in real-world workplace environments with employees aged 30–40 (target demographic).
- User acceptance testing to assess trust, usability, and perceived privacy compliance.
- Collection of feedback on stress detection accuracy and intervention effectiveness.
- Final performance optimization and deployment, ensuring the system meets objectives of real-time operation, personalization, and privacy preservation.

3.11 Technologies and Tools.

1. **Artificial Intelligence (AI)**
    - Hybrid CNN–LSTM model for spatio-temporal stress detection.
    - Lightweight optimization for real-time inference.
2. **Computer Vision**
    - Facial expression recognition using OpenCV.
    - Preprocessing for lighting, alignment, and noise reduction.
3. **Networking & Security**
    - Secure communication with **MQTT + TLS 1.3**.
    - **AES-256 GCM + IV** for centralized DB encryption.
    - Device authentication and role-based access control.
4. **IoT & Analytics**
    - Edge devices integrated with centralized database.
    - Real-time alerts and personalized interventions.
    - Dashboard visualization with stress trends and anonymized reports.

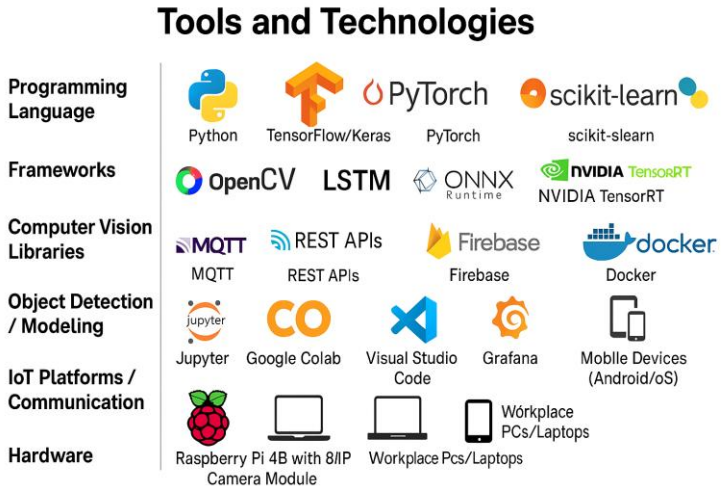

Figure 3 : Technologies and Tools

3.12 Feasibility Study

3.12.1 Technical Feasibility

The proposed research integrates proven technologies including CNN facial recognition, LSTM temporal analysis, and MQTT communication protocols and AES encryption for centralized DataBase. The development team possesses expertise in deep learning, computer vision, mobile application development, ensuring technical viability of the integrated system.

3.12.2 Economic Feasibility

Development Costs:

- Software development and testing infrastructure
- Cloud computing resources for model training
- Edge device procurement for testing and deployment
- User study compensation and data collection expenses

The primarily software-based solution requires minimal hardware investment, making it economically viable for workplace deployment.

3.12.3 Schedule Feasibility

The 12-month development timeline accommodates comprehensive model development, system integration, real-world testing, and validation phases. The agile development methodology enables iterative improvement and stakeholder feedback incorporation throughout the process.

3.12.4 Legal and Ethical Feasibility

- Compliance with workplace privacy regulations and mental health data protection standards
- Informed consent procedures for data collection and system usage
- Ethical approval from institutional review boards
- Clear data ownership and usage agreements with participating organizations

3.13 System Development Methodology

The system development will follow an Agile SCRUM methodology with iterative development cycles, enabling continuous refinement based on user feedback, model evaluation, and real-world testing in workplace environments. Each sprint will deliver functional components that are integrated, tested, and optimized for edge deployment.

3.13.1 Stress Detection Module Development

- **Dataset Preparation**: Collection of facial expression data from employees (ages 30–40) using Raspberry Pi cameras under varied workplace conditions. Annotation of stress intensity levels (0–100 scale) and intervention-response labeling.
- **Model Training**: Development of CNN models for initial feature extraction, followed by integration into CNN–LSTM hybrids for spatio-temporal stress recognition.
- **Performance Optimization**: Conversion of trained models into lightweight TensorFlow Lite / ONNX formats, enabling efficient inference on Raspberry Pi and workplace PCs.

3.13.2 IoT Integration

- **Raspberry Pi Deployment**: Integration of Raspberry Pi 4B with 8MP camera modules for on-device data capture, local preprocessing, and inference.
- **Communication Protocol Implementation**: Secure real-time data transfer using MQTT for stress alerts, REST APIs for system integration, and WebSocket for bidirectional updates.

3.13.3 Recommendation & Personalization Engine

- **Intervention Pool Design**: Building a library of evidence-based interventions (e.g., breathing exercises, mindfulness prompts, music, short breaks).

- **Recommendation Algorithm**: Multi-factor decision-making considering stress severity, context, history, and user preferences.
- **Personalization Mechanisms**: Incorporating collaborative filtering, reinforcement learning, and feedback loops to adapt interventions dynamically over time.

3.13.4 Dashboard & Notification System

- **Dashboard Development**: Grafana-based visualization of stress history, intervention logs, and analytics, updated in real time from detection modules.
- **Notification Optimization**: Delivery of context-aware, multi-modal alerts (push notifications, desktop pop-ups, mobile prompts) to minimize disruption.
- **Integration Testing**: Comprehensive validation of alert accuracy, dashboard synchronization, and intervention records across mobile and desktop platforms.

3.13.5 Security Infrastructure & Database Implementation

- **Centralized DB**: PostgreSQL with AES-256 GCM, unique IVs per record.
- **Key Management**: HSM-based key generation & annual rotation with auto re-encryption.
- **Access Control**: RBAC with tiered permissions, MFA for admins.
- **Compliance & Audit**: GDPR/HIPAA checks, full audit logs, real-time violation alerts.
- **Security Testing**: Regular pentests, vulnerability scans, continuous threat monitoring.

3.14 Project Requirements

3.14.1 Functional Requirements

Real-Time Stress Detection
Detect stress continuously from facial expressions with ≥90% accuracy.
Classify stress into low, moderate, and high levels with response time ≤200ms per frame.

Spatio-Temporal Analysis (Hybrid CNN–LSTM)
CNN extracts spatial facial features (micro-expressions).
LSTM captures temporal stress progression over time.
Update predictions within 1–2 seconds of frame input.

Privacy-Preserving Edge Processing
Process raw video data locally at the device level.
No raw video/images are transmitted outside the device.
Only stress metadata (Employee_ID, stress level, timestamp) is prepared for transmission.

Secure Communication (Networking)
Stress metadata is transmitted from devices to central DB via MQTT over TLS 1.3.
Ensures end-to-end encryption of data in transit.
Only authenticated and authorized devices can send information.

Centralized Database Security
All stress records are stored in a centralized database.
Encrypted with AES-256 in GCM mode.
A unique Initialization Vector (IV) (Device_ID + Counter) is used for each record to guarantee confidentiality and integrity.

Adaptive Intervention & Alerts
Generate personalized interventions (e.g., breathing exercise, mindfulness, music, short break).
Deliver interventions within ≤5 seconds of detecting high stress.
Recommendations adapt based on historical effectiveness, user preferences, and context.

Dashboard & Visualization
Log all stress events, interventions, timestamps, and outcomes.
Provide individual stress trend reports and anonymized organizational analytics (weekly/monthly).
Role-based access ensures HR/managers only see aggregated results.

Alert & Notification System
Deliver alerts via desktop app, mobile notifications, and dashboard.
Must be context-aware (e.g., silent alerts during meetings).
Alerts require acknowledgment to confirm user response.

3.14.2 Non-Functional Requirements

System Performance
The system must process real-time video streams at 20 fps per camera with end-to-end detection and recommendation latency not exceeding 200 ms on Raspberry Pi or workplace PCs.

Reliability and Availability
The system should maintain 99% uptime with fault-tolerant operation. Stress events must not be lost; data loss tolerance must be less than 0.5% under normal operating conditions.

Security and Privacy
All communication (MQTT, REST, WebSocket) must be encrypted using TLS. Role-based access control and audit logs must be implemented to track system usage. Data storage must follow AES-256 encryption standards with auto-deletion protocols.

Resource Efficiency
Models deployed on Raspberry Pi devices must be lightweight (TensorFlow Lite/ONNX optimized) to ensure inference within hardware limits. Devices must operate continuously without overheating or exceeding power constraints.

Usability and User Trust
The interface must be simple and non-intrusive. Employees must understand how data is handled, with transparent explanations of privacy safeguards. Feedback collection should be integrated into the workflow without causing extra stress.

Extensibility
The system should be designed modularly, allowing future integration with third-party HR systems, wearable devices, or advanced intervention libraries without major redesign.

3.15 Expected test cases

1) Test Case - Real-Time Stress Detection and Classification

Objective: Verify that the CNN–LSTM hybrid model accurately detects low, moderate, and high stress levels from facial expressions in real time.
Steps:

1. Record participants (ages 30–40) during varied workplace activities (coding, meetings, multitasking).
2. Annotate ground truth stress levels (low, moderate, high) using psychologist ratings.
3. Stream live video to Raspberry Pi and run local inference.
4. Introduce stress triggers (e.g., workload tasks, time pressure) and recovery phases (short breaks).

Expected Results:

- Stress levels are classified correctly within 1–2 seconds of frame input.
- Predictions remain stable without oscillations from lighting/posture variations.

Measurement:

- Accuracy ≥90%, precision/recall ≥0.90 per class, F1 ≥0.90.

- Latency ≤200 ms end-to-end per frame.
- Fairness: accuracy gap ≤5% across demographics.

2) Test Case - Temporal Stress Progression Tracking

Objective: Validate the system's ability to capture temporal changes (stress build-up and recovery) using LSTM-based sequence modeling.
Steps:

1. Run 30–60 second video clips where stress increases gradually.
2. Feed these sequences into the CNN–LSTM pipeline.
3. Introduce recovery interventions and monitor system response.

Expected Results:

- The system identifies early signs of stress escalation before the participant's peak stress moment.
- Stress decline after intervention is recognized accurately.

Measurement:

- AUC ≥0.92 for escalation prediction.
- Early detection lead time ≥5 s before observed peak.
- Sequence classification latency ≤80 ms per window.

3) Test Case - Personalized Intervention Recommendation

Objective: Ensure that the recommendation engine selects context-aware, effective interventions based on user history and real-time context.
Steps:

1. Configure user profiles with preferences and feedback logs.
2. Simulate high stress during a meeting, coding task, and break time.
3. Deliver interventions (silent breathing prompt, short mindfulness, longer relaxation exercise).

Expected Results:

- Interventions match context (e.g., silent cue in meeting, guided prompt during break).
- Recommendations adapt based on user feedback ("helpful" vs. "not helpful").

Measurement:

- Context compliance ≥95%.
- Intervention acceptance rate ≥80%.
- Personalization improvement: helpfulness score ↑ ≥20% over 2 weeks.

4) Test Case - Real-Time Alerts and Notifications

Objective: Confirm secure, reliable delivery of stress alerts to user devices via MQTT/WebSocket and Firebase push notifications.
Steps:

1. Generate 20 high-stress events and 10 moderate events.
2. Deliver alerts to mobile and desktop applications in real time.
3. Users acknowledge alerts and follow recommendations.

Expected Results:

- Alerts delivered with correct payload (stress level, time, suggested intervention).
- No duplicate or lost alerts.
- User acknowledgment and closure logged in the system.

Measurement:

- Delivery success ≥99%; duplicates ≤1%; dropped alerts = 0.
- Detection → notification latency ≤3 s (median), ≤6 s (p95).
- Acknowledgment within 60 s in ≥90% cases.

5) Test Case - Dashboard Logging and Analytics

Objective: Validate that the dashboard accurately records stress events, interventions, and outcomes for both users and organizational overview.
Steps:

1. Simulate multiple stress detection and intervention cycles across one week.
2. Cross-check dashboard logs against raw system events.
3. Review historical analytics (time-series charts, effectiveness reports).

Expected Results:

- Logs include timestamp, stress level, chosen intervention, and feedback.
- Weekly and monthly summaries are generated automatically.

Measurement:

- Log accuracy ≥99% (match with raw events).
- Report generation time ≤30 s.
- Usability score ≥80/100 (SUS survey).

6) Test Case - Secure Communication Protocol Validation

Objective: Verify that TLS 1.3 with mutual authentication (mTLS) secures all data transmission between edge devices and centralized systems.

Steps:

1. Deploy multiple Raspberry Pi devices with unique X.509 certificates in a test network.
2. Attempt data transmission with valid certificates, expired certificates, and unauthorized devices.
3. Monitor network traffic using packet analyzers to verify encryption integrity.
4. Simulate man-in-the-middle attacks and certificate spoofing attempts.

Expected Results:

- Only devices with valid certificates can establish connections and transmit data.
- All transmitted data remains encrypted and unreadable during network interception.
- Certificate validation failures block unauthorized access attempts.

Measurement:

- Authentication success rate: 100% for valid certificates, 0% for invalid/expired certificates.
- Data encryption verification: 0 readable packets intercepted during transmission.
- Certificate rotation: automated renewal within 24 hours of expiration.

7) Test Case - Centralized Database Encryption and Integrity

Objective: Validate AES-256 GCM encryption with unique initialization vectors (IVs) for all stored stress records.

Steps:

1. Store 1000+ stress records from multiple devices with varying Device_IDs and Counter values.
2. Verify each record uses a unique IV generated from Device_ID + Counter combination.
3. Attempt database access without proper decryption keys.
4. Modify encrypted records and verify integrity detection through GCM authentication tags.

Expected Results:

- All stored records are encrypted and unreadable without proper keys.
- Each record has a unique IV with no collisions across the dataset.
- Any unauthorized modifications trigger integrity violation alerts.

Measurement:

- IV uniqueness: 100% unique IVs across all records.
- Data confidentiality: 0% readable records without decryption keys.
- Integrity protection: 100% detection rate for tampered records.

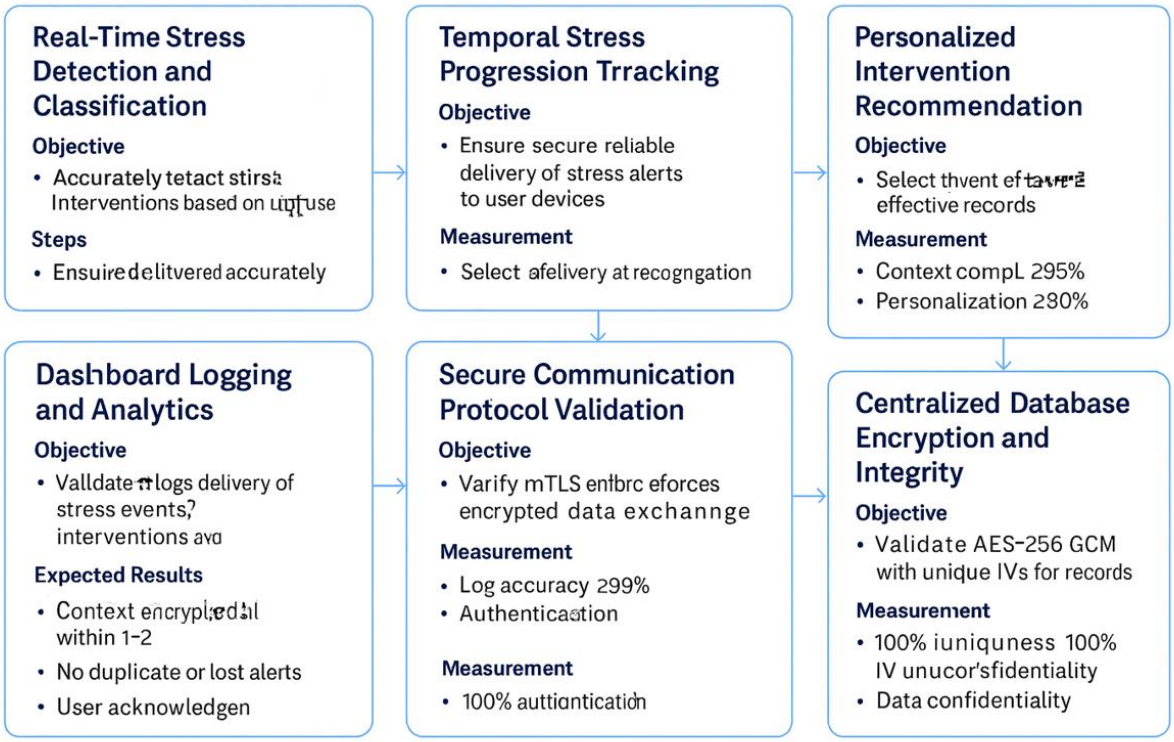

Figure 4 : Expected test cases

3.16 Work Breakdown Structure (WBS)

The following Work Breakdown Structure (WBS) illustrates the major phases and tasks involved in developing the proposed AI-powered real-time stress detection and intervention system for workplace environments. It provides a high-level overview of the project workflow, beginning with dataset preparation and preprocessing, followed by hybrid CNN–LSTM model

development, edge computing and IoT integration, recommendation engine and personalization design, dashboard visualization, and finally testing and validation in real workplace environments. Each phase is broken down into key activities to ensure a structured and systematic approach to achieving the project objectives. This WBS serves as a roadmap for the research, ensuring clarity in task allocation, milestone tracking, and timely project completion while maintaining alignment with privacy, usability, and performance requirements.

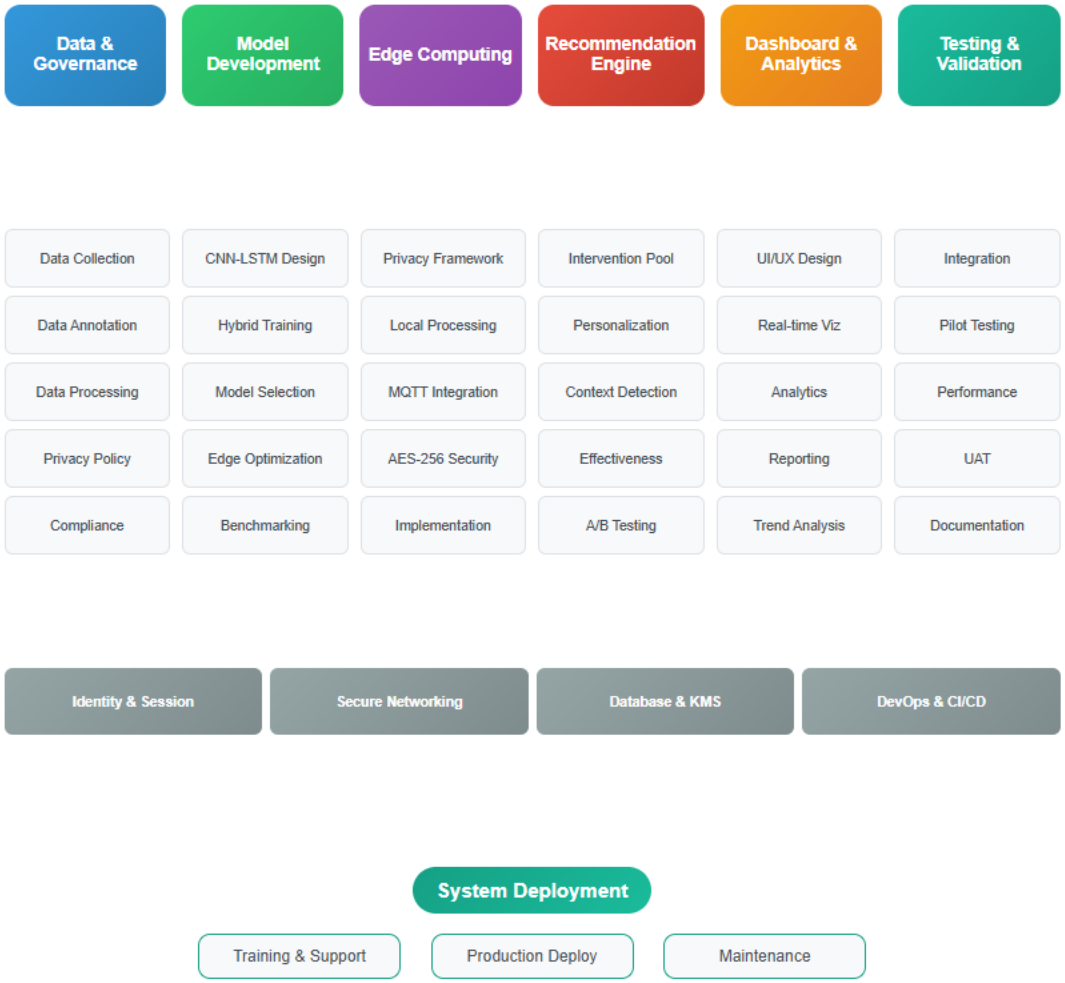

Figure 5 : Work Breakdown Structure

3.17 Gantt Chart

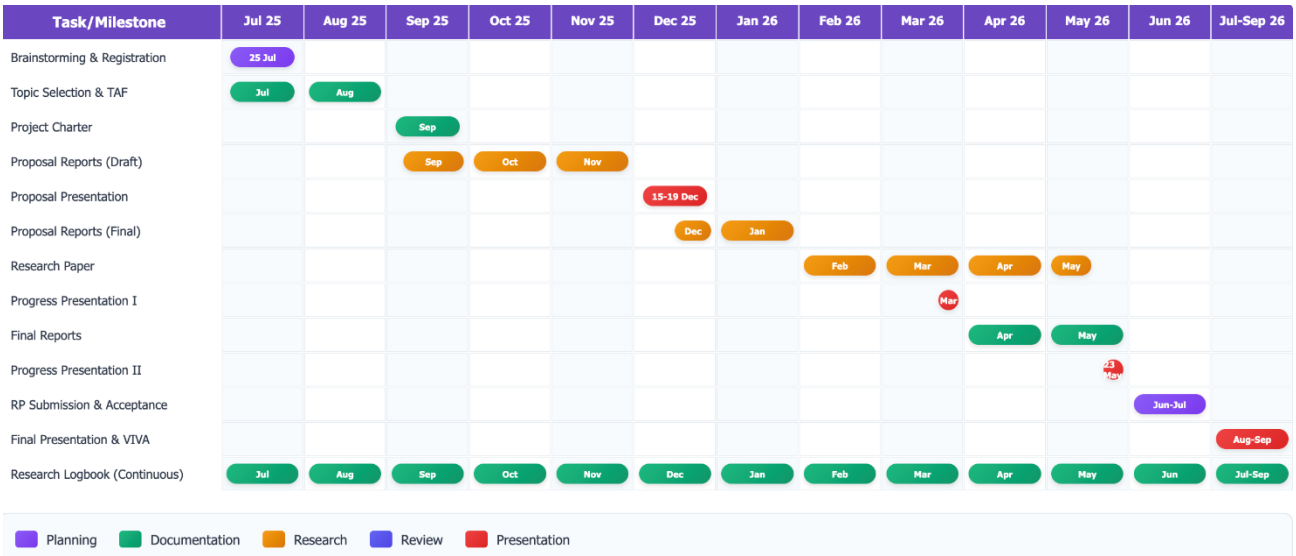

Figure 6 : Gantt Chart

The proposed research study will be executed within a structured 12-month timeline to ensure timely completion and systematic progress. The planning phase will commence in May 2025 and continue until July 2025, covering literature review, stakeholder consultations with IT engineers and psychologists, and requirement analysis. The project initiation phase will run from July 2025 to September 2025, focusing on finalizing the research proposal, defining system specifications, and preparing the custom dataset collection strategy using Raspberry Pi cameras. The implementation phase will begin in September 2025 and extend to February 2026, during which core system modules will be developed, including the hybrid CNN–LSTM model, privacy-preserving edge framework, and intelligent recommendation engine. This will be followed by the testing and evaluation phase from February to April 2026, involving controlled testing, workplace pilot deployment, user acceptance testing, and performance validation. Finally, the documentation and conclusion phase will take place between April and June 2026, covering the preparation of the full research paper, final system documentation, and the project defense presentation. This timeline ensures the research objectives are met systematically, aligning with both academic deadlines and real-world applicability.

4. DESCRIPTION OF PERSONNEL & FACILITIES

This research is supervised by Prof. Anuradha Jayakody, Professor at the Faculty of Computing, Sri Lanka Institute of Information Technology (SLIIT), Malabe, Sri Lanka. She provides high-level academic guidance, ensures research quality, and offers continuous support to align the study with scientific and ethical standards. The project is co-supervised by Mr. Ashvinda Iddamalgoda, who contributes technical expertise in IoT systems, computer networks, and cybersecurity.

The research is carried out by a group of four undergraduate students from the Department of Information Technology, Faculty of Computing, SLIIT. Each member contributes to a distinct subsystem under the umbrella topic "Emotix: a wearable AI system for employee assistant and mental wellness." The distribution of work is as follows:

1. Kariyawasam K.G.A.N (IT22927170) – Secure network-based ai stress detection system for employee, Leads the development of a real-time stress detection system using hybrid CNN-LSTM models for facial expression analysis. The system sends instant alerts with personalized interventions (e.g., breathing exercises, music, or break reminders) and visualizes stress trends on an interactive dashboard.

2. Perera L.W.D.K (IT22601810) – Responsible for developing a workplace assistant that combines biomedical and eye-tracking signals with environmental sensing, integrated with an AI chatbot and voice assistant to provide real-time adaptive feedback and improve employee productivity.

3. **Rathnayaka R.M.H.D (IT22560476)** – Focuses on a self-learning adaptive routine monitoring system using TinyML to learn employee daily patterns, detect deviations, and deliver personalized wellness reminders, adapting routines with employee confirmation for privacy and personalization.

4. Jayarathna S.A.V.P IT22560612 – Designs a secure IoT network communication framework with end-to-end encryption, VPN tunneling, and TPM 2.0 integration. A novel fake robot entry point is introduced to attract attackers, analyze behavior, and trigger AI-driven adaptive defenses for enhanced security.

5. BUDGET AND BUDGET JUSTIFICATION

The budget for this project is carefully allocated to support the successful implementation of the proposed stress detection and intervention system. The primary investment is in Raspberry Pi 4B units with 8MP camera modules (Rs. 5,500 each), which will serve as edge devices for real-time video capture, preprocessing, and local inference. These devices are critical for ensuring privacy-preserving data collection and real-time monitoring in workplace environments.

Additionally, workplace PCs/laptops will be leveraged for training and testing, while cloud services such as Google Colab or equivalent compute support will be used for advanced model training and fine-tuning during the research phase. To facilitate real-time communication and notifications, Firebase services (Rs. 1,000) will be utilized for push notifications and temporary data synchronization between devices and mobile applications.

This allocation ensures a cost-effective setup that balances hardware for edge deployment with cloud-based computational resources for training. The budget prioritizes scalability, real-time responsiveness, and strict privacy preservation.

Budget Breakdown

Table 2 : Budget Justification

Resource Type	**Unit Cost (LKR)**	**Quantity**	**Total (LKR)**
Raspberry Pi 4B + 8MP Camera Module	Rs. 5,500	1	Rs.15,500
Firebase Services (Push Notifications, DB)	Rs. 1,000	1	Rs. 1,000
Miscellaneous (Cables, Mounts, Power Supply, Storage)	Rs.500	1	Rs. 500
Total			**Rs.17,000**

This budget is justified as it ensures real-time stress monitoring, privacy-preserving edge deployment, and scalable cloud training support while remaining cost-efficient and practical for workplace testing.

Reference List

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